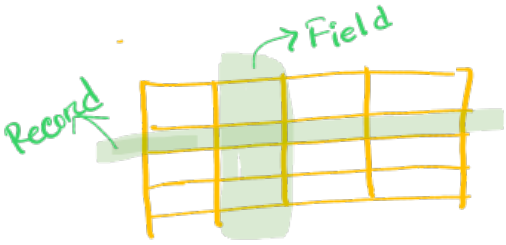
Piece of Data → Record.
Each item in Record → Field.



Product → Record
code, name, unit price → Fields.
Each record have field of the same type.
Fields can be unique. → identifies data
Absence of value → Null

Relational Data Model

Primary Key: Unique field ⇒ identifies data.
Projection → operation to extract a column.

Cartesian product, difference, intersection, selection, Projection, Division
Union
join

Projection → vertically
Selection → horizontally.

Foreign Key →

Entity → recognizable object.
Normalization → Dividing the table to prevent data conflicts.

The table that has rows with two or more values before it is divided is called the unnormalized form

The table that results from a division like this is called the first normal form.

a table in which values in other columns are determined when the primary key is determined → 2nd Normal Form.

A table that does not allow any nonprimarykey to determine values in other columns...is called the third normal form

!! practice qstns at page 76.

Features of SQL:
1. Creation of tables
2. Input and retrieval of data
3. Management of users

SQL

```
SELECT product_name
FROM product
WHERE unit_price>=200
```

or * → will select product code || name || unit price - all columns.

```
SELECT *
FROM product
WHERE product_name='apple';
```

→ Retrieve

ProCod	ProNa	UnitP
103	APPLE	120G

Express the unknown part using %, like this...

```
SELECT *
FROM product
WHERE product_name LIKE '%n';
```

→ MELON, LEMON — both

– they end with N

TO SORT products in order of ascending price, add a statement like ORDER BY unit SELECT *
FROM product
WHERE unit_price<200
ORDER BY unit_price;price.

→ 103 APPLE 120G
102 STRAWBERRY 150G

In the SELECT phrase, use AVG (column name) to obtain the average of each row

SELECT AVG(unit_price)
FROM product;

Aggregate Functions in SQL

Function	Description
COUNT(*)	Obtains the number of rows
COUNT(column_name)	Obtains the number of times the column is not null
COUNT(DISTINCT column_name)	Obtains the number of distinct values in the column

SUM(column_name)	Obtains the sum of the column's values in all rows
AVG(column_name)	Obtains the average of the column's values in all rows
MAX(column_name)	Obtains the maximum value of the column
MIN(column_name)	Obtains the minimum value of the column

In order to join tables, SQL requires a condition that...the primary key is equal to the foreign key which refers to the primary key.

Join tables by placing a comma between them.
If the same column name appears in multiple tables, just specify it as TABLE_NAME.COLUMN_AME.

```
SELECT  sales.report_code, date,
sales.export_destination_code,
export_destination_name, sales_statement.product_code,
product_name, unit_price, quantity
FROM sales, sales_statement, product, export_destination
WHERE sales.report_code = sales_statement.report_code
AND
sales_statement.product_code = product.product_code
AND
export_destination.export_destination_code =
sales.export_destination_code
```

CREATE TABLE → make a table.

```
CREATE TABLE product
```

(
product_code int NOT NULL,
product_name varchar(255),
unit_price int,
PRIMARY KEY(product_code)
);
You must specify the
primary key, as well.
I used the product code
as a primary key.*
We've also set the
datatype of each
column. You can see
that product and unit_
price are integers (int).
Varchar means that the
database expects text,
and (255) limits the
product_name to 255
characters

INSERT → add data.

INSERT INTO product
(product_code,product_name,unit_price)
VALUES (101,'melon',800);

When two or more concurrent transactions yield the same result as if they were performed at separate times, that order of processing is referred to as serializable.

optimizer → function in charge of

optimization of queries

Nested loop, Sort Merge, Hash

3 tier client/server system → presentation layer, logic layer, data layer.